

Predicting World Cup Outcomes with Elo Ratings and Machine Learning

Leakage-safe feature engineering, chronological evaluation and Monte Carlo simulation



Abhirup - Sripal [25200786 - 25214828]

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FRAME

BUILD

EVALUATE

1. PROBLEM, DATA & FEATURES

QUESTION / MOTIVATION

International team strength evolves over time, and tournament outcomes compound uncertainty across many matches. The project therefore focuses on probabilities built only from information available before kick-off.

RESEARCH QUESTION

Can Elo ratings and leakage-safe recent-form features produce useful match probabilities for World Cup simulation?

DATA AT A GLANCE

49,287 raw
49,215 validated
25,157 modern era

21,712 training matches 2000–2022
3,445 test matches 2023–2026

LEAKAGE-SAFE FEATURES

RECENT FORM

Previous 5 matches
Goals scored + conceded

TOURNAMENT CONTEXT

Competition importance
Neutral-venue indicator

PRE-MATCH ELO

Dynamic team strength
Captured before kick-off

PERSISTENT MATCH ID

Deterministic match_id
Safe one-to-one joins

FORECAST INTEGRITY

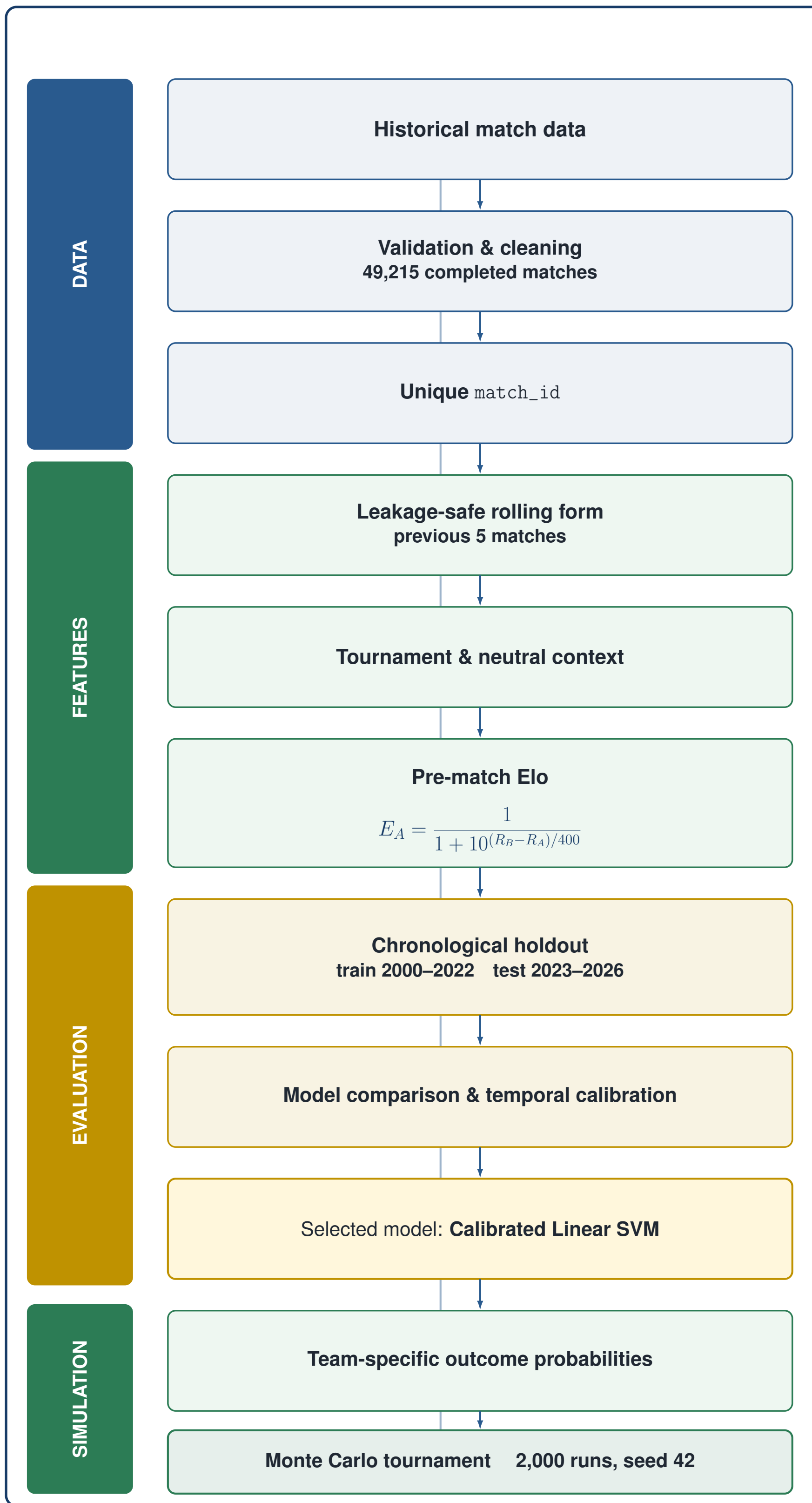
- NO TARGET LEAKAGE**
Current match excluded from its own rolling form.
- SAME-DAY SAFE**
Same pre-date form and Elo state for same-day appearances.
- PRE-MATCH ELO ONLY**
Ratings recorded before the result updates team strength.
- SAFE JOINS**
One-to-one match_id joins prevent row multiplication.
- TEMPORAL HOLDOUT**
No date overlap; standardisation fit on training data only.

FROZEN EXTERNAL-TEST STATE

Classifier training end: 30 Dec 2022 | Team-state end: 31 Mar 2026 | Gap: 72 days
World Cup matches in model data: 0 | Retraining: 0 | Reselection: 0

2. METHOD

FROM HISTORY TO PROBABILITIES



WHY PROBABILITY METRICS?

Log loss, Brier score and calibration evaluate the full away-win / draw / home-win probability vector, not only the argmax class.

FROZEN COMMIT 11 STATE → 2026 WORLD CUP BACKTEST

State end 31 Mar 2026 | 104 unseen matches | no retraining, retuning or model reselection.

3. RESULTS

MAIN RESULT

Calibrated Linear SVM

59.62% Accuracy
0.445 Macro F1
0.892 Log loss
0.525 Brier

Selected for the lowest chronological test-set log loss among the evaluated models.

MODEL COMPARISON

Model	Accuracy	Macro F1	Log loss	Brier
Calibrated Linear SVM	59.62%	0.445	0.892	0.525
Balanced HistGradientBoosting	58.35%	0.429	0.926	0.543
Class-prior baseline	46.85%	0.213	1.056	0.637
Baseline → selected model	Accuracy 46.85% → 59.62%		Log loss 1.056 → 0.892	

MODEL BEHAVIOUR

Confusion matrix

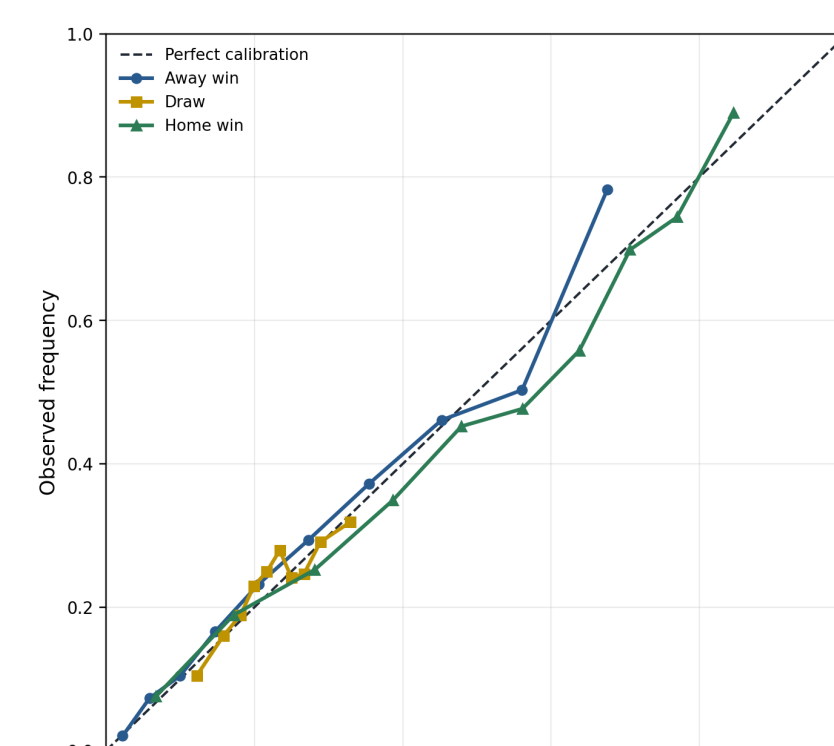
Rows = actual; columns = predicted

	PREDICTED		
	Away	Draw	Home
Away	646	12	378
Draw	285	9	501
Home	209	6	1399

Lower Higher count

Home wins dominate correct hard classifications; draws are rarely the top-probability class.

Calibration



Reliability curves compare forecast probability with observed outcome frequency on the chronological holdout.

2026 WORLD CUP EXTERNAL BACKTEST

104 Matches
5,000 bootstrap resamples
60.58% Accuracy
95% CI 50.96%–70.19%
0.884 Log loss
95% CI 0.786–0.984
0.524 Brier
95% CI 0.452–0.597

Draws: 23.08% observed | 21.50% mean predicted | 0.00% argmax recall

Frozen pre-tournament evaluation; bootstrap seed 2026. Low hard-class draw recall does not imply negligible draw probability.

4A. KEY FINDINGS & LIMITATIONS

3 KEY FINDINGS

- Leakage-safe evaluation matters**
Chronological testing gives a more realistic forecast estimate than random splitting.
- The calibrated SVM improves on baseline**
Accuracy: 46.85% → 59.62% | Log loss: 1.056 → 0.892
- Frozen external validation holds up**
104 unseen matches: 60.58% accuracy | 0.884 log loss | 0.524 Brier.

LIMITATIONS / NEXT STEPS

LIMITATIONS

72-DAY FRESHNESS GAP

Frozen state ends 31 Mar 2026; later friendlies, injuries and squad changes are excluded.

PLAYER / SQUAD STATE

No April–June squad selections, player availability or tactical updates.

TOURNAMENT STRUCTURE

Bracket is an approximate demonstration structure.

NEXT STEPS

SQUAD STRENGTH

Add player availability and squad-strength features.

BOOKMAKER COMPARISON

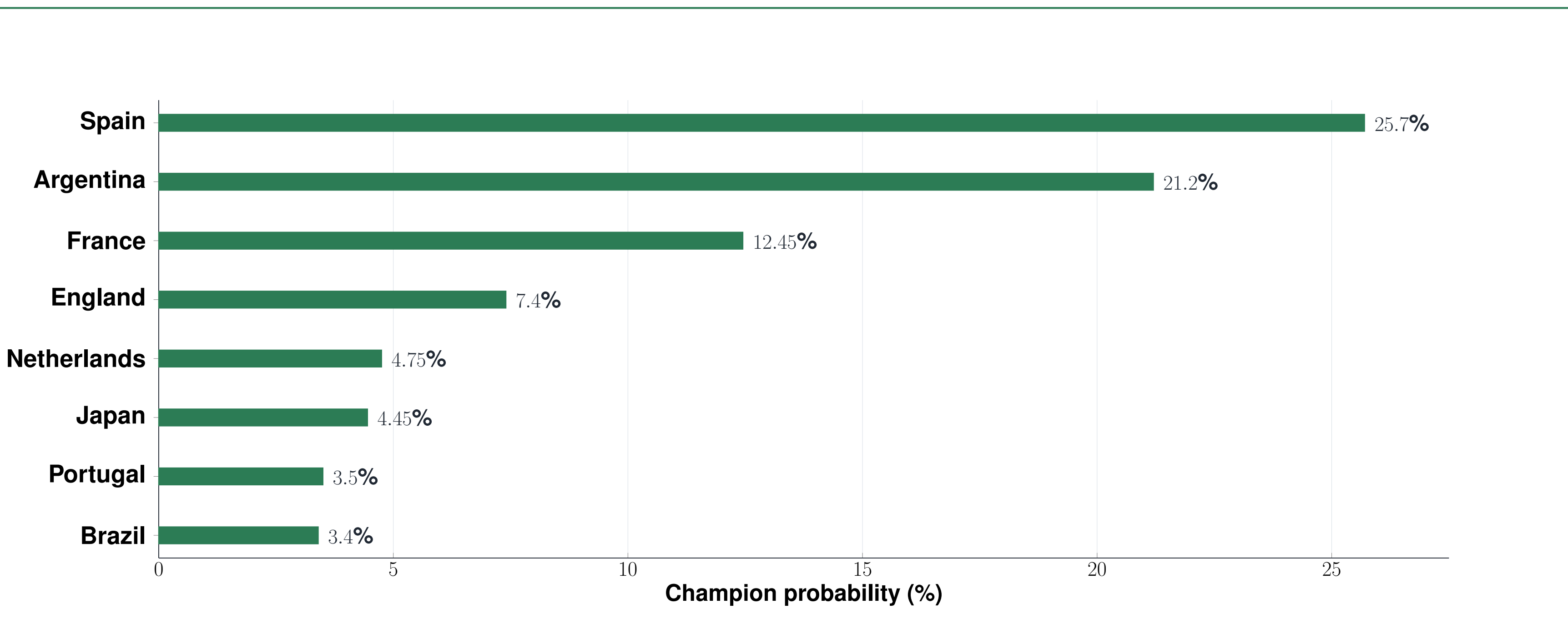
Compare calibrated probabilities with bookmaker benchmarks.

EXACT TOURNAMENT LOGIC

Integrate the exact schedule and bracket logic.

4B. PRE-TOURNAMENT 2026 WORLD CUP SIMULATION

ACTUAL 48-TEAM FIELD: CHAMPION PROBABILITIES



2,000
SIMULATIONS

2026
SEED

48
ACTUAL TEAM FIELD

3-way
MATCH PROBABILITIES

INTERPRETATION

Frozen simulation ranked Spain first (25.70%) and Argentina second (21.20%). Spain later won; model quality is judged on all 104 match probabilities, not the champion pick alone.

Retrospective comparison; no World Cup outcome entered training or reselection.

SELECTED REFERENCES

Baboota & Kaur (2019)
Predictive analysis and modelling of football results.
Constantinou & Fenton (2012)
Probabilistic football forecast evaluation.
Ding, Guo & Xu (2026)
WC2026-Agents benchmark; 104-match ground truth.

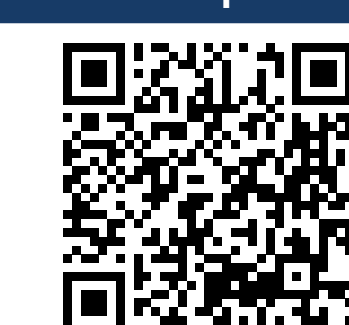
REPRODUCIBILITY

Source of truth: Model outputs + World Cup backtest CSVs.

Actual results remain separate from model-development data; provenance is documented in WORLD_CUP_2026_SOURCES.md. The frozen model was not retrained or reselected.

CODE & RESULTS

GitHub repository



Scan for code, methodology and reproducible results